

Supervised ML / Deep Learning Presentation Summary
Source: Supervised_ML_DL_Python_Presentation.pdf
Relevant Slide: page_10_Image65.jpg (ML pipeline diagram)

ML Pipeline Vocabulary and Concepts:

1. Data Collection and Exploration

- collect samples
- identify ranges
- compute histograms
- identify anomalies

2. Data Cleaning and Preprocessing

- handle missing values
- correct/remove errors
- normalize data
- partition datasets (train/test split)
- dimensionality reduction

3. Feature Engineering

- create interaction terms
- add polynomial features
- select important features
- transform features

4. Model Training and Validation

- train models on training data
- tune hyperparameters
- validate models on held-out validation data

5. Model Evaluation and Selection

- evaluate models on test data
- compare metrics
- select best performing model

How this maps to ML_DL_Summer_School_2026.ipynb:

- train_test_split and scaling = Data Cleaning and Preprocessing
- logistic regression training = Model Training and Validation
- ROC curve and classification report = Model Evaluation and Selection
- discussion of Glucose, BMI, Age = Feature Engineering / feature importance